

AWS Cloud Practitioner

A one-day, hands-on introduction to Amazon Web Services aligned to the AWS Certified Cloud Practitioner (CLF-C02) exam

AWS Cloud Practitioner Duration 1 Day Format 70% Labs / 30% Theory

Course Overview

This course provides a comprehensive introduction to Amazon Web Services (AWS). It covers the fundamental concepts of cloud computing, core AWS services, security best practices, and cost optimization strategies. By the end of this course, learners will have a solid understanding of AWS and be able to design, deploy, and manage cloud-based applications on the AWS platform.

Length: 1 day (9:00 AM – 4:00 PM) **Format:** ~70% hands-on labs / ~30% theory (225 min labs, 105 min theory, plus breaks and lunch) **Level:** Introductory — no prior AWS experience assumed **Environment:** AWS Management Console in a shared AWS account, Region **us-east-2 (Ohio)**. No IDE or local tooling required.

Course Objectives

Upon completion of this course, learners will be able to:

- Explain the core concepts of cloud computing and the benefits of using AWS
- Identify and describe key AWS services, including EC2, S3, VPC, IAM, and more
- Understand the AWS global infrastructure and regions
- Create and manage AWS resources using the AWS Management Console and CLI
- Implement basic security best practices on AWS
- Optimize AWS costs using various strategies
- Prepare for the AWS Certified Cloud Practitioner exam

Course Outline

Module	Topics
1. Introduction to Cloud Computing	What is cloud computing · benefits · types (public, private, hybrid, multi-cloud) · IaaS / PaaS / SaaS
2. Core AWS Services	2A Security, Identity & Compliance: IAM, security groups & NACLs, KMS · 2B Networking & Content Delivery: VPC, Route 53, CloudFront · 2C Compute: EC2 (instance types, AMIs, key pairs, security groups), Lambda · 2D Storage: S3, EBS · 2E Databases: RDS, DynamoDB
3. AWS Global Infrastructure	Regions and Availability Zones · edge locations · Global Accelerator
4. Billing and Cost Optimization	Pricing models · Cost Explorer · Reserved Instances and Savings Plans · rightsizing · Auto Scaling
5. Best Practices and Troubleshooting	Building secure and reliable applications · common troubleshooting techniques · Well-Architected Framework · exam preparation

Module 2 is taught in five short blocks, each immediately followed by its lab, so teaching content always precedes hands-on work.

Daily Schedule

Time	Block	Min
9:00 – 9:05	Welcome & environment check	5
9:05 – 9:17	Module 1: Introduction to Cloud Computing	12
9:17 – 9:27	Module 2A: Security, Identity & Compliance — IAM, KMS	10
9:27 – 9:57	 Lab 1: Configuring IAM Roles and Policies	30
9:57 – 10:07	Module 2B: Networking & Content Delivery — VPC, SG/NACL, Route 53, CloudFront	10
10:07 – 10:37	 Lab 2: Configuring Security Groups and Network ACLs	30
10:37 – 10:52	 Break	15
10:52 – 11:00	Module 2C: Compute — EC2	8
11:00 – 11:25	 Lab 3: Creating EC2 Instances	25
11:25 – 11:30	Module 2C: Deploying applications on EC2	5
11:30 – 12:00	 Lab 4: Deploying a Web Application on EC2	30
12:00 – 1:00	 Lunch	60
1:00 – 1:08	Module 2D: Storage — S3 & EBS	8
1:08 – 1:33	 Lab 5: Using S3 for Object Storage	25
1:33 – 1:40	Module 2C: Serverless — Lambda	7
1:40 – 2:05	 Lab 6: Implementing Serverless Functions with Lambda	25
2:05 – 2:13	Module 2E: Databases — RDS & DynamoDB	8
2:13 – 2:18	 Lab 7 Part A: Launch the RDS instance	5
2:18 – 2:33	 Break (RDS provisions meanwhile)	15
2:33 – 2:58	 Lab 7 Part B: Connect to and explore RDS	25
2:58 – 3:06	Module 3: AWS Global Infrastructure	8
3:06 – 3:18	Module 4: Billing and Cost Optimization	12
3:18 – 3:48	 Lab 8: Implementing Cost Optimization Strategies + cleanup	30
3:48 – 4:00	Module 5: Best Practices, Troubleshooting & Exam Preparation	12

Totals: labs 225 min · theory 105 min · breaks/lunch 90 min.

Hands-on Labs

Every lab runs in the console with a per-student `userN-` naming prefix, and each lab builds on the previous one — the VPC from Lab 2 hosts the servers from Labs 3–4 and the database from Lab 7; the IAM role from Lab 1 is proven from EC2 in Labs 3 and 5. Lab 8 ends with an ordered cleanup checklist.

Lab	Title	Duration	Core task
1	Configuring IAM Roles and Policies	30 min	Group + user with a managed policy; test least privilege; EC2 role
2	Configuring Security Groups and Network ACLs	30 min	VPC (2 AZs) via wizard; web and DB security groups; one NACL Deny rule
3	Creating EC2 Instances	25 min	Key pair; launch Amazon Linux 2023 with the role; connect via Instance Connect
4	Deploying a Web Application on EC2	30 min	Install Apache + page; create AMI; launch second server with user data
5	Using S3 for Object Storage	25 min	Versioned bucket; static website with bucket policy; read from EC2 via role
6	Implementing Serverless Functions with Lambda	25 min	Python function; test + logs; Function URL
7	Setting Up a Relational Database with RDS	5 + 25 min	MySQL in the VPC behind the DB security group; connect from EC2; find backups/Multi-AZ
8	Implementing Cost Optimization Strategies	30 min	Pricing Calculator comparison; Budget; rightsize an instance; cleanup

Course Materials

```

aws_cloud_pract/
├── README.md                # This file – overview and schedule
├── build-pdfs.sh            # Regenerates everything under pdf/ from the
sources (~1 min)
├── tools/print.mjs          # Chrome DevTools PDF printer used by build-
pdfs.sh
├── presentations/
│   └── aws_cloud_practitioner.html # Reveal.js deck for the whole day (single
file, 58 slides)
├── pdf/                    # PDF copies of every deliverable (rebuild
with ./build-pdfs.sh)
│   ├── aws_cloud_practitioner_slides.pdf # Deck, one page per slide
│   ├── aws_cloud_practitioner_lab_manual.pdf # All 8 labs in one student handout
│   └── README.pdf
└── lab-exercises/
    ├── lab01/README.md      # IAM roles and policies
    ├── lab02/README.md      # Security groups and network ACLs (VPC
build)
    ├── lab03/README.md      # Creating EC2 instances
    ├── lab04/README.md      # Deploying a web application on EC2
    ├── lab05/README.md      # S3 object storage
    ├── lab06/README.md      # Lambda serverless functions
    ├── lab07/README.md      # RDS relational database
    └── lab08/README.md      # Cost optimization + cleanup

```

Every deliverable exists in its source format (HTML / Markdown) and as a PDF. After editing any source, run (needs Google Chrome, Node 22+, pandoc and poppler):

```
./build-pdfs.sh
```

Getting Started

For Students

1. Get your username (userN), sign-in URL and password from the instructor.
2. Begin with [Lab 1 — Configuring IAM Roles and Policies](#) when the instructor reaches the first lab callout.
3. Complete Labs 1–8 in order; each lab lists what to keep for later labs and what to clean up.

Exam Alignment (CLF-C02)

Exam domain	Weight	Where it is covered
1. Cloud Concepts	24%	Module 1, Module 3, Module 5 (Well-Architected)
2. Security and Compliance	30%	Module 2A, Lab 1, Lab 2, Module 5
3. Cloud Technology and Services	34%	Modules 2B–2E, Module 3, Labs 2–7
4. Billing, Pricing and Support	12%	Module 4, Lab 8, Module 5 (support plans)

The exam is 65 questions in 90 minutes, passing score 700/1000. Module 5 closes with eight practice questions and a study plan.